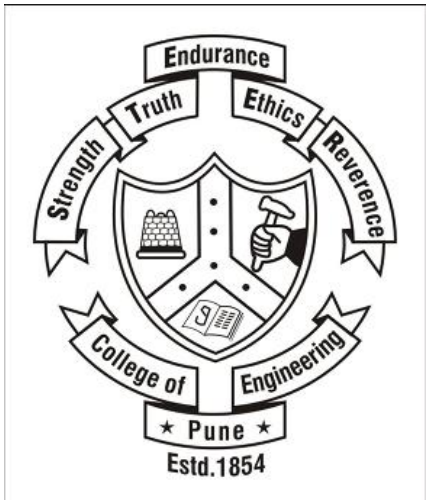
COLLEGE OF ENGINEERING, PUNE



B.Tech Project

Computer Vision Based Robotic Arm

Guided by Dr Shantipal S Ohol

Department of Mechanical Engineering

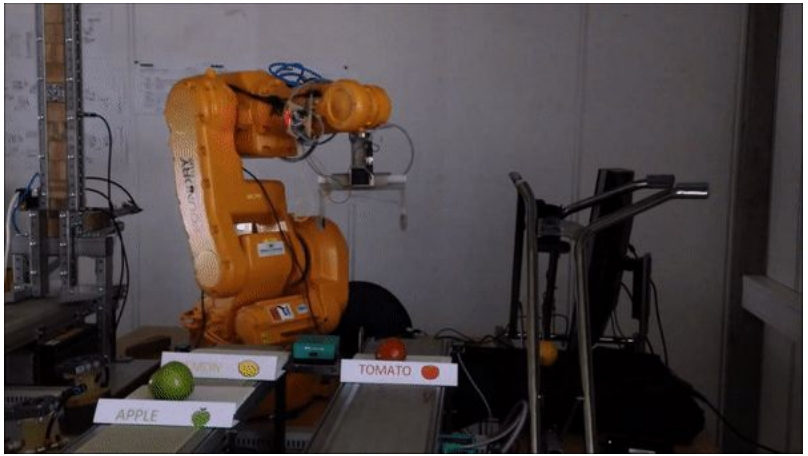
AIM – Programming Robotic Arm for Fruit Sorting Operation

GROUP NO.38 MECH

1. Anshul Chandekar
2. Tejas Ukey
3. Ankush Ukey
4. Mayuresh Kshirsagar

INTRODUCTION & PROBLEM - STATEMENT





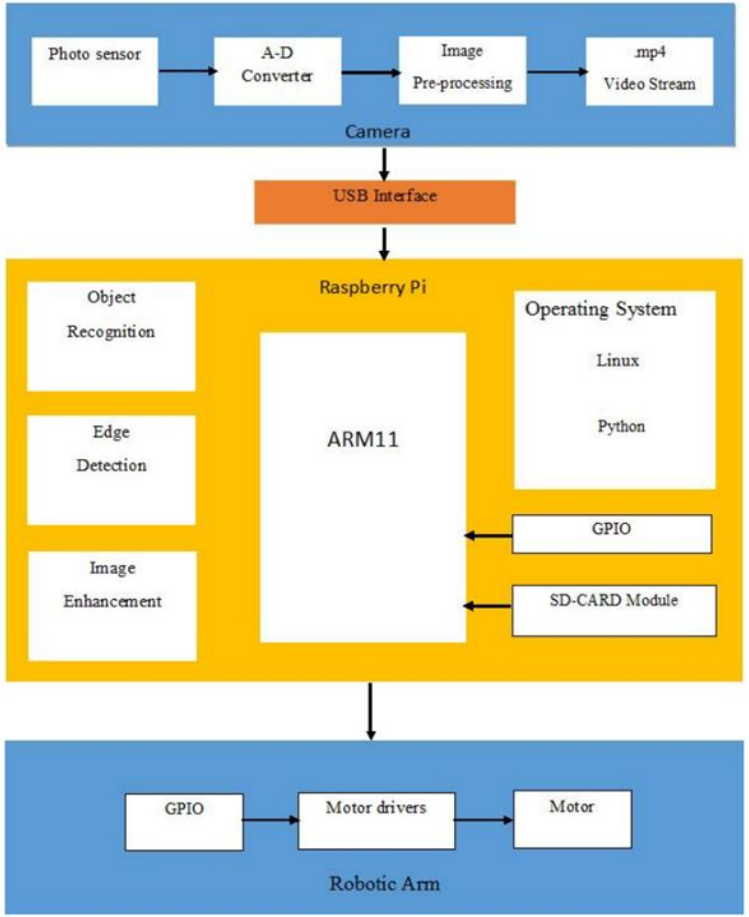
LITERATURE SURVEY

- A working system is presented in the paper using an industrial articulation robot manipulator and a computer vision named KHOROS A stereo camera system was used for guidance and sensing. The greatest advantage of the system at hand is that it does not require a priori knowledge. All work conducted is in line with the principle of first demonstrating the relevant objects, situations and/or actions, and then learning from those.

- A technique for finding the relative spatial pose between a robotic end-effector and a target object to be grasped without a priori knowledge of the spatial relationship between the camera and the robot is presented.

- A working system was implemented using an adaptive closed-loop controller with visual guidance for grasping using a monocular approach .The camera configuration used was a robot-mounted camera (eye-in-hand configuration). The target objects were simple rectangular prisms with one graspable dimension. For a static object task, the success rate was 96%, whereas for moving objects was 80%.

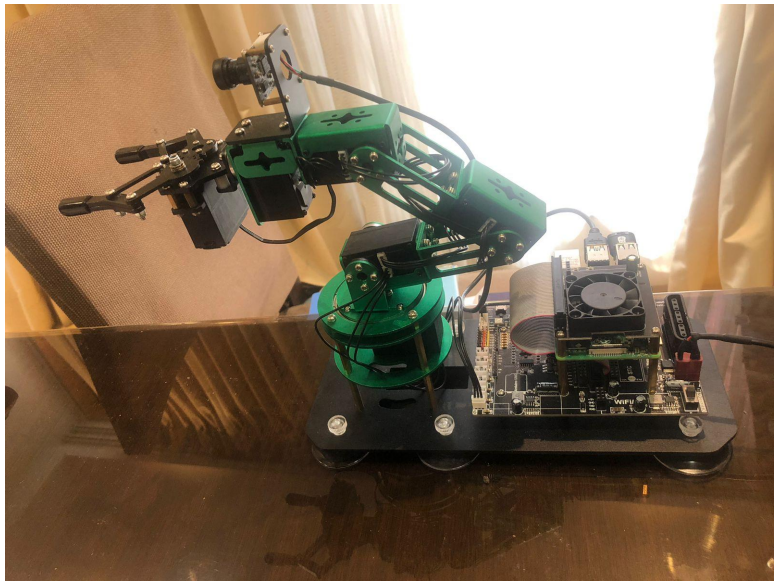
WORKING


Photo sensor
A-D Converter
Image Pre-processing
.mp4 Video Stream
Camera
USB Interface
Raspberry Pi
Object Recognition
Edge Detection
Image Enhancement
ARM11
Operating System
Linux
Python
GPIO
SD-CARD Module
GPIO
Motor drivers
Motor
Robotic Arm

CHALLENGES AND DRAWBACKS

- ADVANTAGES
- 1. better human computer interface
- 2. relatively cheap raspberry pi
- DISADVANTAGES
- 1. deploying deep learning model on raspberry pi
- 2. lighting conditions

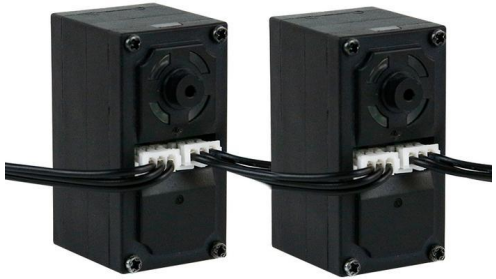
ROBOTIC ARM





IMPORTANT PARTS







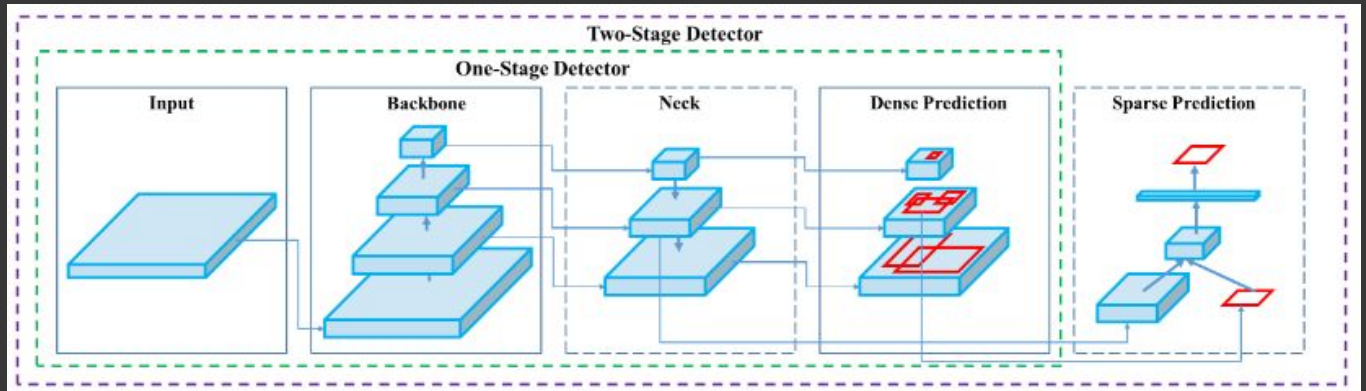






1. *Servo motors: serial bus smart servo motor*
2. *RASPBERRY PI:-*
3. *Aluminum alloy body*
4. *Rgb cooling HAT*
5. *USB cable*
6. *Ethernet cable*
7. *Connecting wires*
8. *Keyboard & Mouse*

OBJECT DETECTION – YOLO V5



Input: { Image, Patches, Image Pyramid, … }

Backbone: { VGG16 [68], ResNet-50 [26], ResNeXt-101 [86], Darknet53 [63], … }

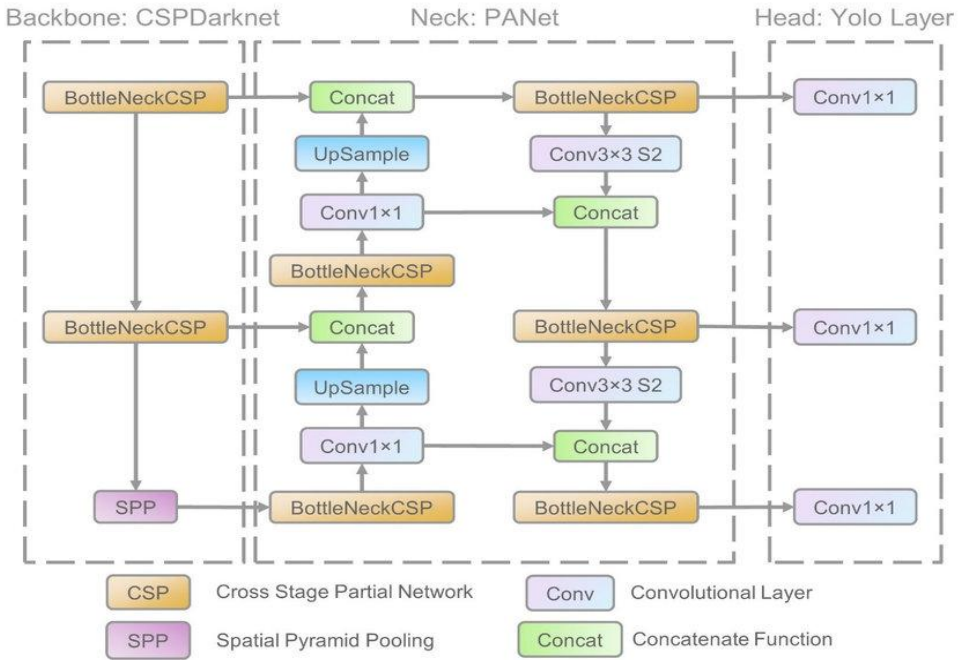
Neck: { FPN [44], PANet [49], Bi-FPN [77], … }

Head:

Dense Prediction: { RPN [64], YOLO [61, 62, 63], SSD [50], RetinaNet [45], FCOS [78], … }

Sparse Prediction: { Faster R-CNN [64], R-FCN [9], … }

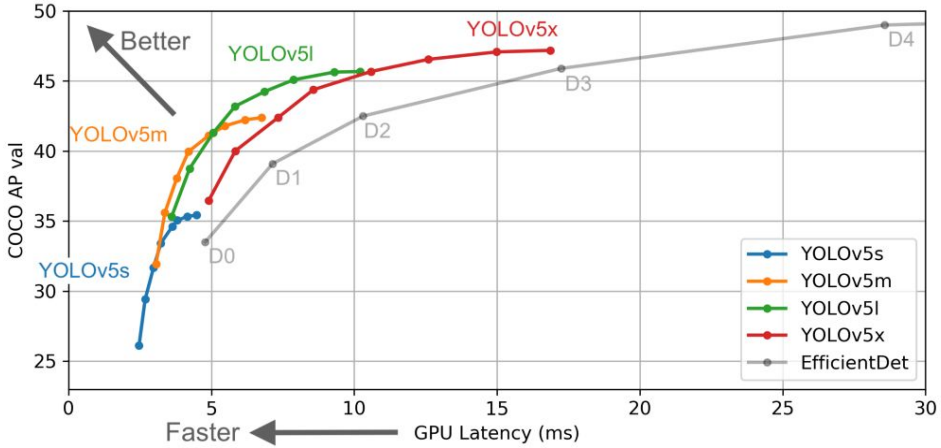
YOLO ARCHITECTURE

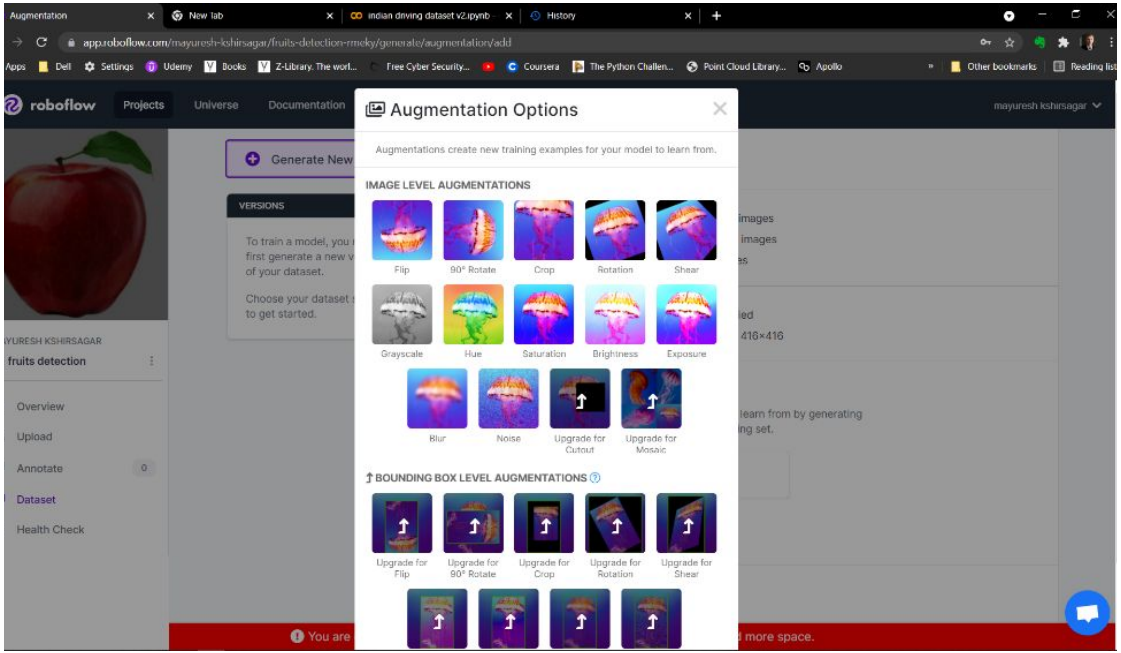


TYPES OF YOLO ARCHITECTURES



Small	Medium	Large	XLarge
YOLOv5s	YOLOv5m	YOLOv5l	YOLOv5x
14 MB_{FP16} 2.0 ms_{V100} 37.2 mAP_{COCO}	41 MB_{FP16} 2.7 ms_{V100} 44.5 mAP_{COCO}	90 MB_{FP16} 3.8 ms_{V100} 48.2 mAP_{COCO}	168 MB_{FP16} 6.1 ms_{V100} 50.4 mAP_{COCO}





TRAINING DATASET AND ROBOFLOW SERVICE

TRAINING AND RESULTS

Colaboratory output of gpu utilization and mean average precision per epoch on total of 40 epochs

```
           Class     Images     Labels          P          R     mAP@.5 mAP@.5:.95: 100% 1/1 [00:03<00:00,  3.06s/it]
             all         58        115     0.0193     0.0748      0.011    0.00219

     Epoch   gpu_mem       box       obj       cls    labels  img_size
     37/39     8.95G   0.07721   0.03669   0.03332        52       416: 100% 8/8 [00:30<00:00,  3.76s/it]
           Class     Images     Labels          P          R     mAP@.5 mAP@.5:.95: 100% 1/1 [00:03<00:00,  3.07s/it]
             all         58        115     0.0214     0.0677      0.011    0.00211

     Epoch   gpu_mem       box       obj       cls    labels  img_size
     38/39     8.95G   0.07579   0.03355   0.03349        39       416: 100% 8/8 [00:30<00:00,  3.76s/it]
           Class     Images     Labels          P          R     mAP@.5 mAP@.5:.95: 100% 1/1 [00:03<00:00,  3.06s/it]
             all         58        115     0.0213     0.0748     0.0156     0.0028

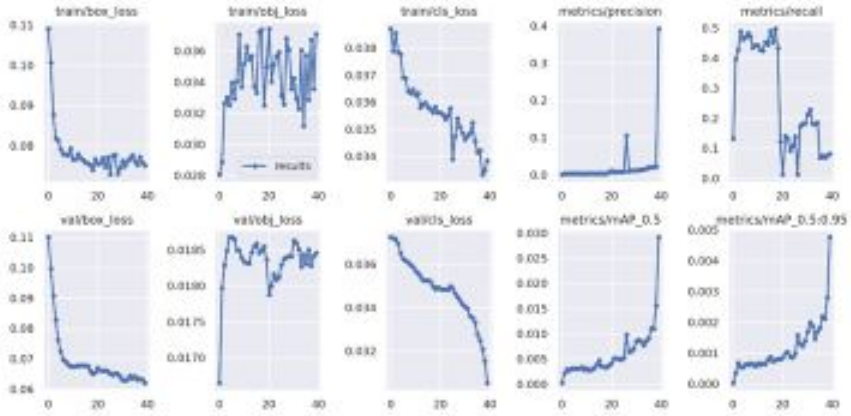
     Epoch   gpu_mem       box       obj       cls    labels  img_size
     39/39     8.95G   0.07506   0.03706   0.03384        57       416: 100% 8/8 [00:30<00:00,  3.76s/it]
           Class     Images     Labels          P          R     mAP@.5 mAP@.5:.95: 100% 1/1 [00:03<00:00,  3.05s/it]
             all         58        115      0.392     0.0816     0.0292    0.00476

40 epochs completed in 0.391 hours.
Optimizer stripped from runs/train/yolov5l_results/weights/last.pt, 93.7MB
Optimizer stripped from runs/train/yolov5l_results/weights/best.pt, 93.7MB

Validating runs/train/yolov5l_results/weights/best.pt...
Fusing layers...
Model Summary: 392 layers, 46611336 parameters, 0 gradients, 114.1 GFLOPs
           Class     Images     Labels          P          R     mAP@.5 mAP@.5:.95: 100% 1/1 [00:03<00:00,  3.97s/it]
             all         58        115      0.392     0.0816      0.029    0.00473
           apple         58         49      0.176      0.245     0.0814     0.0132
          banana         58         38          0          0    0.00163   0.000278
          orange         58         28          1          0    0.00398   0.000765
Results saved to runs/train/yolov5l_results
CPU times: user 8.18 s, sys: 1.28 s, total: 9.46 s
Wall time: 24min 9s
```

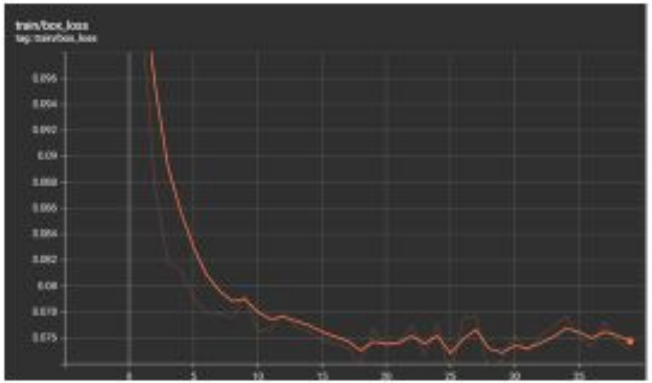
TENSORBOARD GRAPHS

Training and Validation graphs over 40 epochs





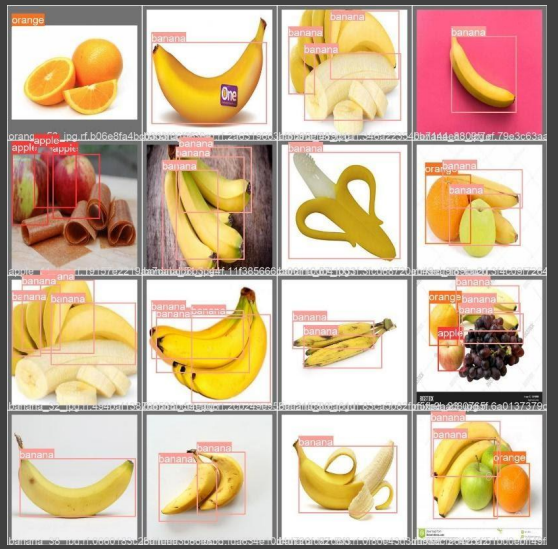
Mean average Precision Vs Number of epochs



Loss Vs Number of epochs.

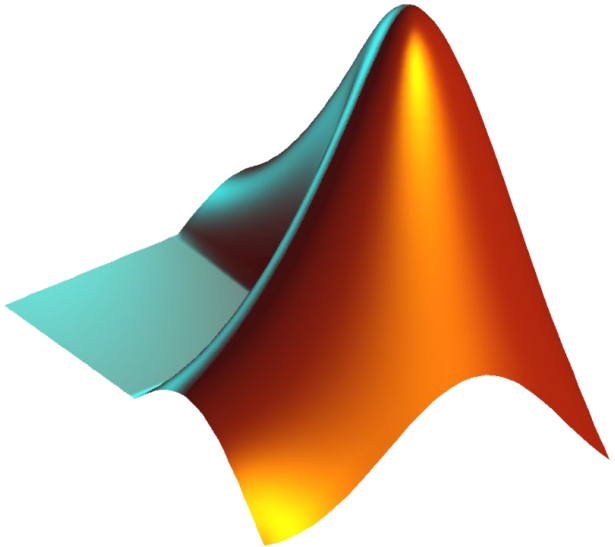


REAL TIME IMAGE PROCESSING

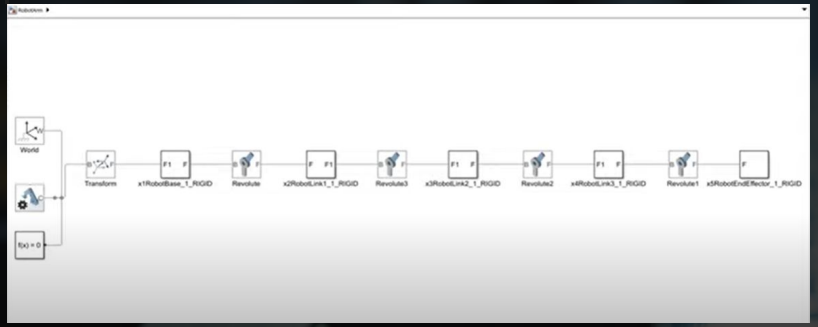


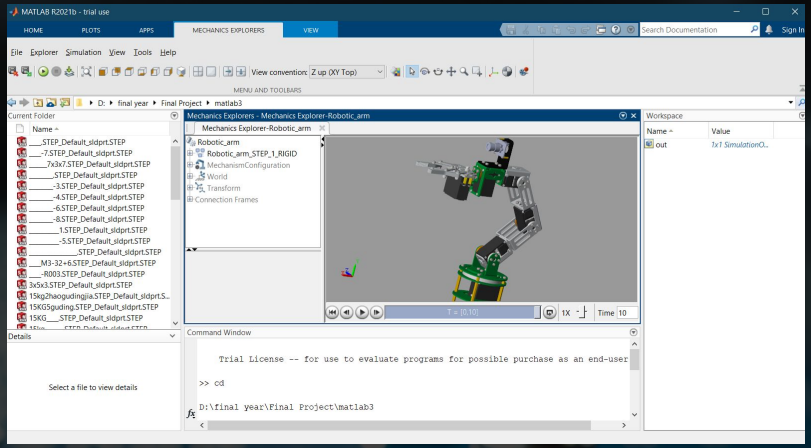
MATLAB

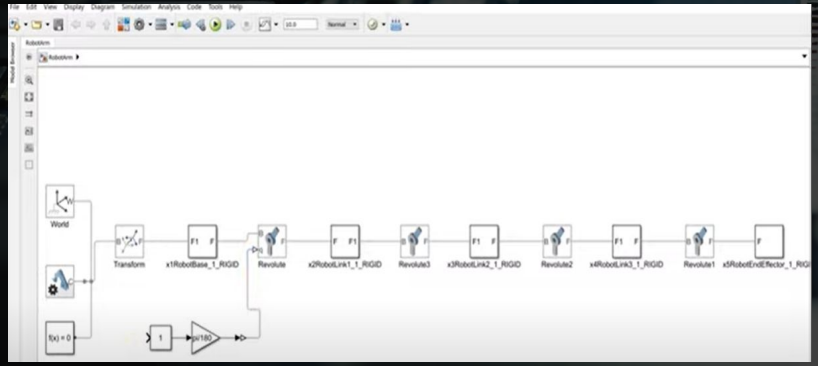
- Stands for MATrix LABoratory
- Programming language that expresses matrix and array mathematics directly
- It integrates computation, visualization, and programming environment

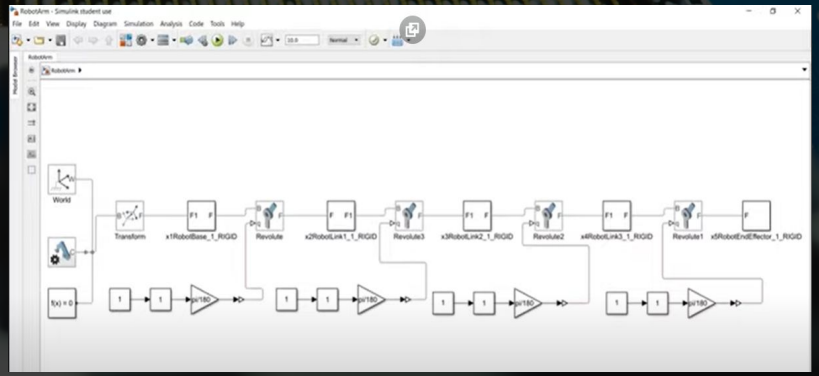


MATLAB - SIMULINK









ACHIEVEMENTS



1. Trained object detection model for our specific task



2. Simulated the robot in MATLAB

1. CO-ORDINATE TRANSFORMATION

2. PATH PLANNING

FUTURE CHALLENGES

Reference

1. Official yolov5 github repository - https://github.com/ultralytics/yolov5 , ultralytics .
2. Yolov4 paper - https://arxiv.org/abs/2004.10934 , Alexey Bochkovskiy, Chien-Yao Wang, Hong-Yuan Mark Liao.
3. Learning to Recognize and Grasp Objects - https://link.springer.com/article/10.1023/A:1008874725911 , Josef Pauli.
4. Relative spatial pose estimation for autonomous grasping - https://www.spiedigitallibrary.org/journals/optical-engineering/volume-36/issue-12/0000/Relative-spatial-pose-estimation-for-autonomous-grasping/10.1117/1.601586.short?SSO=1 , Steve Roach, Michael Magee.
5. Issues and experimental results in vision-guided robotic grasping of static or moving objects - https://www.emerald.com/insight/content/doi/10.1108/01439919810204748/full/html , Nikolaos Papanikolopoulos , Christopher E. Smith .

Thank You